Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Constants

R = 0.08206 L·atm/mol·K
R = 8.314 J/mol·K
R = 62.36 L·torr/mol·K
e = 1.602×10^{-19} C
N_A = 6.022×10^{23} mol^{-1}
k = 1.38×10^{-23} J/K
h = 6.626×10^{-34} J·s
c = 3.00×10^{8} m/s
$\mathcal{R}$ = 2.178×10^{-18} J
m_e = 9.11×10^{-31} kg

Water data

T_{mp} = 0°C
T_{bp} = 100°C
C_{ice} = 2.09 J/g K
C_{water} = 4.184 J/g K
C_{steam} = 2.03 J/g K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/m
K_b = 0.512 °C/m

Vapor Pressure (torrs) of water at selected temperatures

15 °C	12.79
20 °C	17.54
25 °C	23.77
30 °C	31.84

Selected acid-base data

pKa HClO = 7.53

pKa $CH_2ClCOOH$ = 2.85

pKa_1 H_2CO_3 = 6.37

pKa_2 H_2CO_3 = 10.25

pKb NH_3 = 4.74

Selected Equilibrium Constants

K_{sp} $Zn(OH)_2$ = 4.5×10^{-17}

K_{sp} $Pb(OH)_2$ = 2.8×10^{-16}

K_{sp} $Mn(OH)_2$ = 4.5×10^{-14}

K_w = 1.0×10^{-14}

Conversions

1 atm = 760 torr
= 101.335 kPa
= 14.7 psi
1 bar = 10^5 Pa
1 cal = 4.184 J
1 L•atm = 101.325 J
1 eV = 1.602×10^{-19} J
1 lb = 453.6 g
1 ton = 2000 lbs
1 tonne = 1000 kg

Selected Thermodynamic Data

ΔH_f $CO_{2(g)}$ = -393.51 kJ

ΔH_f $CO_{(g)}$ = -110.53 kJ

ΔH_f $H_2O_{(g)}$ = -241.82 kJ

Standard Reduction Potentials

$Mn^{2+} + 2e^- \rightarrow Mn(s)$ $\mathcal{E}^o$ = - 1.03 V